Chapter 11, Problem 40P

Problem

Design a problem to help other students better understand apparent power and power factor.

Step-by-step solution

Step 1 of 1

Design a problem to help other students better understand apparent power and power factor.

Although there are many ways to work this problem, this is an example based on the same kind of problem asked in the third edition.

Problem

A load consisting of induction motors is drawing 80 kW from a 220-V, 60 Hz power line at a pf of 0.72 lagging. Find the capacitance of a capacitor required to raise the pf to 0.92.

Solution

$$pf1 = 0.72 = \cos\theta_1 \longrightarrow \theta_1 = 43.94^0$$

$$pf2 = 0.92 = \cos\theta_2 \longrightarrow \theta_2 = 23.07^0$$

$$C = \frac{P(\tan\theta_1 - \tan\theta_2)}{\omega V_{rms}^2} = \frac{80x10^3(0.9637 - 0.4259)}{2\pi x60x(220)^2} = \underline{2.4\text{ mF}}$$

{Again, we need to note that this capacitor will be exposed to a peak voltage of 311.08V and must be rated to at least this level, preferably higher!}